BreatheEasy

Your asthma action plan, matched to tomorrow's air.

DSH Hacks V2 · AI × Healthcare
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4.8 million
U.S. children (6.7%) have asthma; 47% had an attack in the past year (CDC, 2023 NHIS)

6th worst
Houston-Pasadena metro for ozone in the U.S. (American Lung Association, State of the Air 2026)

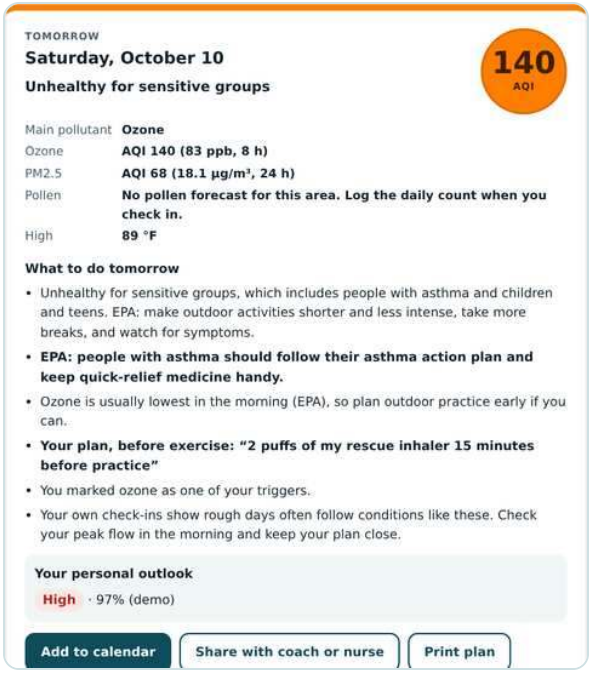
43.3 days/yr
weighted unhealthy ozone days, Harris County, 2022–2024 (ALA)

The problem

Doctors give families a paper **asthma action plan** (green, yellow, red zones). On bad-ozone days EPA tells people with asthma to "follow their action plan and keep quick-relief medicine handy". But the plan sits in a drawer and the forecast lives in another app, so they rarely meet the night before, when there's still time to move practice indoors or take the pre-exercise medicine the doctor already prescribed.

What BreatheEasy does

- **Tomorrow's AQI, the EPA way.** Hourly ozone and PM2.5 forecasts (Open-Meteo / Copernicus CAMS) turned into a daily U.S. AQI with EPA's breakpoints and averaging rules (8-h ozone, 24-h PM2.5, 2024 PM2.5 update).
- **Guidance in your doctor's words.** EPA's advice for that level beside the text you copied from your own plan. It never invents medicines or doses.
- **20-second check-in.** Peak flow and symptoms → your zone (NHLBI cut-offs: 80% / 50% of personal best), that zone's steps, and a Call 911 button on danger signs.
- **On-device AI that explains itself.** A logistic-regression model trains in the browser on your own check-ins, learns which conditions (ozone, PM2.5, pollen, heat, humidity, yesterday) come before your rough days, explains each prediction feature by feature, and reports held-out accuracy vs. a naive baseline. Silent until it has 14 days of data.
- **Share, calendar, print**: a note for a coach or school nurse, an .ics event, or a printout. **English and Spanish.** Mobile-first and accessible.
- **Private**: no account, no server, no tracking; data stays in the browser with one-tap export and delete.



Tomorrow card on a real forecast (Sugar Land, TX, Oct 10, 2026): EPA-method AQI 140; Open-Meteo's own peak hourly AQI was 143. Personal outlook shown here uses the labelled synthetic demo data.

How it's built and tested

Static HTML/CSS/JavaScript (no framework, no keys, runs on any phone). Modules for EPA AQI math, forecast processing, NHLBI zones, guidance selection, the model, i18n and calendar export. **44 unit tests** (including EPA's worked ozone example and every breakpoint edge) and **58 headless-Chromium checks** (zones, 911 alert, model, export, Spanish, accessibility, 390 px layout, API failure). On a clearly labelled **synthetic** 75-day demo, the model recovers the planted pattern (pollen and ozone on top) with **84% accuracy on 19 held-out days vs. a 68% baseline**; that proves the pipeline, not clinical performance.

Honest limits and next steps

Forecasts are model estimates, not monitor readings. There is no free U.S. pollen forecast, so users log the Houston Health Department count (labelled as logged, never as a forecast). Next: pilot with families and a school nurse, AirNow Ozone Action Day alerts, reminders and an offline installable app.

Not medical advice; not a medical device. BreatheEasy helps you follow the plan your doctor wrote. Trouble walking or talking from shortness of breath, or blue lips or fingernails: call 911.

Sources: CDC Most Recent National Asthma Data (2023 NHIS) · American Lung Association, State of the Air 2026 (Texas) · NHLBI Asthma Action Plan (2020) · U.S. EPA, 40 CFR Part 58 App. G and Air Quality Guide for Ozone · Open-Meteo Air Quality API (CAMS, CC BY 4.0). Built with help from an AI coding assistant; every statistic was checked against its source.